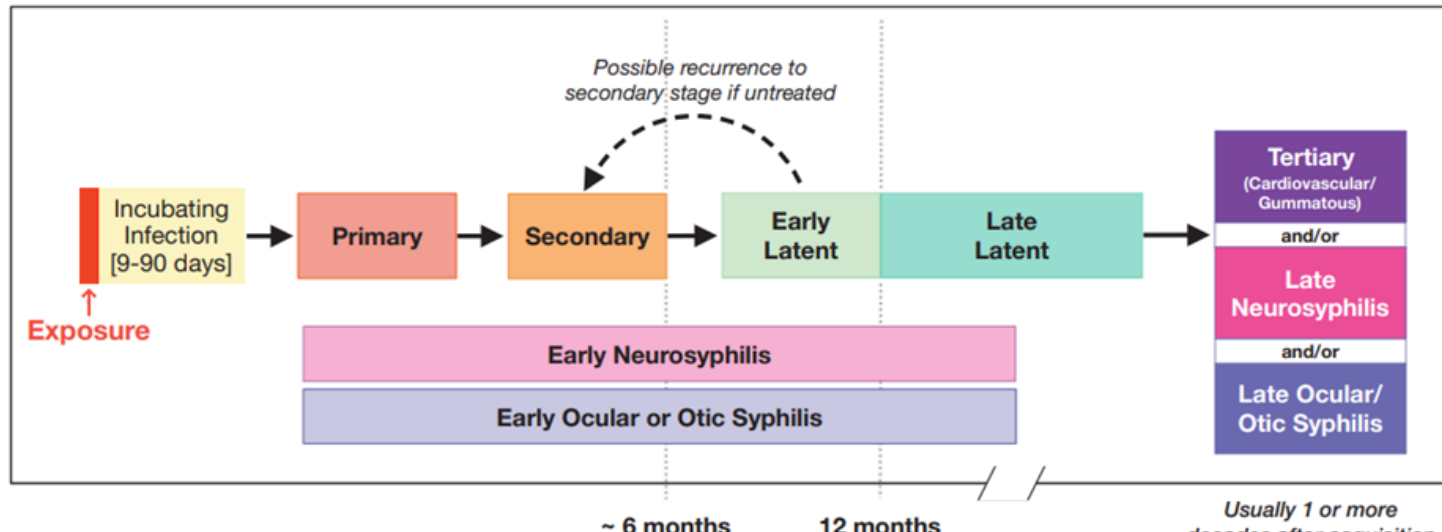




## Syphilis on the Rise

The rate of syphilis has grown almost every year since 2001; it increased 28.6% between 2020 and 2021. In 2021, the US had 176,713 cases of reported syphilis; that is a 74% increase since 2017. During that time, there was also a 203% increase in cases of congenital syphilis.

Syphilis is caused by a spirochete, *Treponema pallidum*. Inoculation occurs from contact with infected secretions from a chancre, mucous patch, or condyloma lata. Clinical manifestation depends on the stage of the disease.



Note. From *The Natural History of Untreated Syphilis* [Infographic], by the New York City Department of Health and Mental Hygiene, and the New York City STD Prevention Training Center, 2019, *The diagnosis and management of syphilis: An update and review* ([https://www.nycdoh.org/x/Syphilis\\_Monograph\\_2019\\_NYC\\_PTC\\_NYC\\_DOHMH.pdf](https://www.nycdoh.org/x/Syphilis_Monograph_2019_NYC_PTC_NYC_DOHMH.pdf)).

## Early Syphilis: The Most Infectious Stage

**Primary syphilis:** Painless ulcer (chancre) at the site of inoculation. Multiple lesions may occur. Lesions may go unnoticed and heal spontaneously without treatment.

**Secondary syphilis:** Disseminated, systemic infection that can last several weeks or months without treatment. Even if asymptomatic, without treatment it may relapse, especially within the first year. Therefore, it is considered infectious (early latent below).



Note. From *Secondary syphilis resembling erythema multiforme* [Photograph], by Chinmoy Bhate et. al., 2010, *International Journal of Dermatology* (<https://onlinelibrary.wiley.com/doi/10.1111/j.1365-4632.2009.04390.x>).

- Non-specific symptoms: fever, headaches, malaise, myalgias
- Diffuse nontender adenopathy, but involvement of epitrochlear nodes is suggestive of diagnosis.
- Diffuse symmetric macular or papular rash involving entire trunk, extremities classically involving palms and soles. Lesions can also be pustular.
- Condyloma lata
- Mucous patches
- Alopecia: “moth-eaten” alopecia
- Hepatitis: high alk phos with relatively normal transaminases

## Latent Syphilis: Infected but Asymptomatic

**Early latent:** Infection occurred within the previous 12 months. There is a high risk of transmission, including maternal-fetal transmission. Patient with new serologic evidence of syphilis or a history of syphilis with 4-fold or greater increase from baseline in nontreponemal titer.

**Late latent:** Infection occurred > 12 months ago. The risk of transmission is low.

## Late (Tertiary) Syphilis

Late syphilis occurs in about 20% of patients with untreated syphilis. It can occur even in patients without symptomatic early syphilis.

Symptoms are variable and depend on the sites affected. Common sites include:

- Cardiovascular-aortitis
- CNS: General paresis and tabes dorsalis
- Gummas

**Central nervous system (CNS), ocular, and otic involvement can occur at any stage of infection. The patient may either remain asymptomatic or manifest a variety of neurologic, psychiatric, visual, auditory, and vestibular signs and symptoms.**



Note. From *A large cutaneous lesion* [Photograph], by Centers for Disease Control and Prevention, n.d., Centers for Disease Control and Prevention (<https://phil.cdc.gov/Details.aspx?pid=17837>).

## Diagnosis

Interpretation of serologic tests depends on the presence or absence of clinical disease, the patient’s prior history of syphilis, prior treatment, and the individual’s immune status.

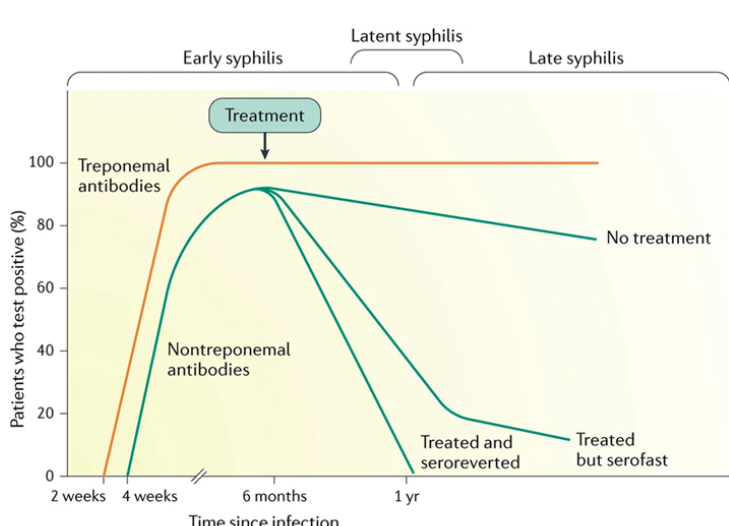
**Serologic testing:** It may be negative during early infection. If the patient is symptomatic and there is high clinical suspicion, you can treat presumptively and repeat testing in 2 to 4 weeks.

**Treponemal-specific test:** Detects antibodies directed against specific treponemal antigens. It is more specific than nontreponemal tests. It does not distinguish between active and previously-treated syphilis. It usually remains positive for life. Some patients who are treated early during primary syphilis may become seronegative after a few years.

Limitations:

- May be negative in very early infection.
- May be positive due to other spirochete infections.

**Nontreponemal test (RPR or VDRL):** Non-specific. Positive results can be seen in other acute infections, autoimmune disease, immunizations, etc. .



Note. From *Serological response to primary and secondary syphilis* [Infographic], by Nature Reviews, n.d., Nature Reviews: Disease Primers (<https://www.nature.com/articles/nrdp201773/figures/6>).

Limitations:

- May be negative in very early infection, with prior-treated syphilis, late or late latent syphilis where nontreponemal tests have become nonreactive over time even without treatment, or prozone phenomenon (very high titers cannot agglutinate without dilution).

## Treatment

Penicillin is the treatment of choice.

**Empiric treatment for patients with:**

- Classic chancre
- Known recent exposure

**Primary, secondary, and early latent:**

- Single-dose penicillin G benzathine 2.4 million units IM
- Alternative treatment: Doxycycline 100 mg bid x 14 days, or Ceftriaxone 1 g daily IM or IV for 10 to 14 days

**Tertiary syphilis or late latent syphilis:**

- If no CNS involvement: Penicillin G benzathine 2.4 million units IM weekly for 3 weeks
- Alternative treatment: Doxycycline 100 mg bid x 4 weeks, or Ceftriaxone 2 g daily IM or IV for 10 to 14 days

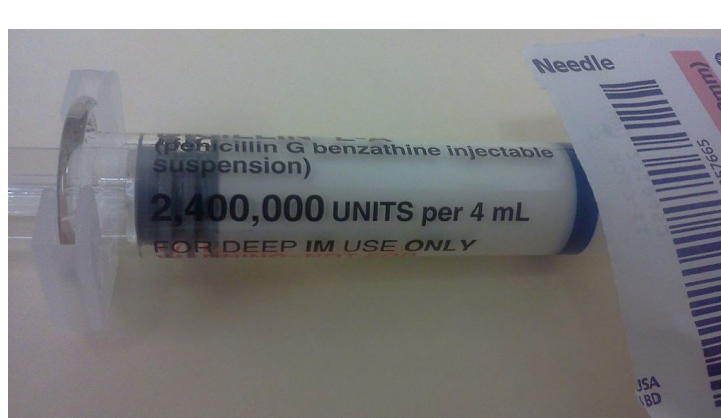
**CNS (early neurosyphilis or late neurosyphilis), ocular, and otic syphilis can occur during any stage: IV penicillin is the preferred treatment. Alternatives include IM penicillin daily, plus probenecid for 10-14 days or ceftriaxone IV for 10-14 days.**

**Response to treatment and monitoring after treatment:**

- Clinical resolution of rash or ulcer.
- Serologic testing with nontreponemal test (RPR) at 6 and 12 months (and 24 months if following up response after treatment for late syphilis)—should see a 4-fold or greater decline in nontreponemal titer (RPR)—(i.e., change from 1:16 to 1:4).
- Most patients will serorevert (clinical cure) after successful treatment over time, especially if treatment was initiated early.
- Causes for nontreponemal titers to decline less than 4-fold include treatment failure, reinfection, undiagnosed neurosyphilis, or very low initial titers (<1:8).

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Note. From *Benzylpenicillin, 2,400,000 units, for deep intramuscular injection* [Photograph], by DanZ, 2014, Wikimedia Commons ([https://commons.wikimedia.org/wiki/Category:Penicillin\\_antibiotics#/media/File:Bicillin\\_L-A\\_\(Benzylpenicillin\).jpg](https://commons.wikimedia.org/wiki/Category:Penicillin_antibiotics#/media/File:Bicillin_L-A_(Benzylpenicillin).jpg)).